NON-MEASUREMENT SENSITIVE

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31 May 1983

MILITARY SPECIFICATION

TRUCK, FIREFIGHTING; TRACTOR-TRAILER TYPE, 100-FOOT AERIAL LADDER, DED

This specification is approved for use by the Naval Facilities Engineering Command (YD), Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a diesel engine powered tractor drawn, trailer type aerial ladder unit for use essentially in rescue operation and combating structural fires.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4210

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SPECIFICATIONS

FEDERAL

TT-E-529 - Enamel, Alkyd, Semi-Gloss.

MILITARY

MIL-T-704 - Treatment and Painting of Material.
MIL-I-45208 - Inspection System Requirements.

STANDARDS

FEDERAL

FED-STD-297 - Rustproofing of Commercial (Nontactical) Vehicles.
FED-STD-595 - Colors.

MILITARY

MIL-STD-281 - Automobiles, Trucks, Truck-Tractors, Trailers and Trailer Dollies, Preservation and Packaging of.
MIL-STD-889 - Dissimilar Metals.
MIL-STD-1223 - Nontactical Wheeled Vehicles Treatment, Painting, Identification Marking, and Data Plate Standards.

(Copies of specifications, standards, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other Government documents and publications. The following other Government documents and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DEPARTMENT OF TRANSPORTATION (DoT)

Federal Motor Vehicle Safety Standards and Regulations.
Federal Motor Carrier Safety Regulations.

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

2.3 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise

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specified, the issues of the documents not listed in the DODISS shall be the issue of the non-Government documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A36/A36M - Steel Structural.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

ENVIRONMENTAL PROTECTION AGENCY (EPA)

Control of Air Pollution from New Motor Vehicles and New Motor Vehicle Engines: Certification and Test Procedures.
Interstate Motor Carrier Noise Emission Standards.
Motor Vehicle Air Pollution Standards.

(Application for copies should be addressed to the Public Affairs Office, Environmental Protection Agency, Rockville, MD 20852; or CFR, Title 40 should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

1901 - Automotive Fire Apparatus.
1904 - Fire Department Aerial Ladder and Elevating Platforms.
1931 - Fire Department Ground Ladders.

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269.)

SOCIETY OF AUTOMOTIVE ENGINEERS, INC. (SAE)

SAE Handbook.

(Application for copies should be addressed to the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

TIRE AND RIM ASSOCIATION, INC. (TRA)

TRA Yearbook.

(Application for copies should be addressed to the Tire and Rim Association, Inc., 3200 West Market Street, Akron, OH 44313.)

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(Non-Government standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.4 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The truck shall conform to the provisions of NFPA 1901, applicable to apparatus equipped with an aerial ladder, with the modifications and additional requirements specified herein. The truck shall consist of a four-wheel tractor with a suitable fifth wheel mounted upon the rear of the chassis to carry the forward end of the aerial ladder trailer unit and a semitrailer suitable for the mounting and transporting of the aerial ladder. The tractor shall be diesel-engine-driven, front wheel steerable, and rear wheel driven. A tiller and operator's seat shall be provided to steer the rear wheels of the trailer unit. A 100-foot hydraulic powered metal aerial ladder shall be mounted on the trailer unit.

3.2 First article. When specified (see 6.1), the contractor shall furnish a truck for first article inspection and approval (see 4.2.1 and 6.3).

3.2.1 Inspection system requirements. Quality assurance inspection and tests necessary to substantiate product conformance to contract specifications shall be in accordance with MIL-I-45208.

3.3 Standard truck, components, and accessories. Except as specified herein, the truck, components, and accessories shall be standard or optional items which meet or exceed the requirements of this specification. All items shall be as represented and rated in the manufacturer's sales information, including special or mounted equipment. Sales information shall be limited to specifications and technical material, identical to that furnished to the authorized company representative for selection of fire truck models and components.

3.4 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and

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reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification unless otherwise specified.

3.4.1 Dissimilar metals. Unless protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MIL-STD-889.

3.5 Environmental conditions. The truck shall be capable of starting and of satisfactory performance in any ambient temperature from +125 degrees Fahrenheit (°F), while exposed to maximum solar radiation, to -25°F. The truck shall meet the performance requirements at any elevation for sea level to 2,000 feet at the ambient temperatures specified.

3.6 Performance. The fully equipped truck, loaded as specified in 3.7.4, shall be capable of the following:

a. Ascend, both in forward and reverse gears, a grade of not less than 30 percent.
b. Accelerate on a dry, level, paved road from a standing start to a speed of 35 miles per hour (mph) in not more than 30 seconds.
c. From a steady speed of 15 mph, in direct drive, on a dry, level, paved road, the truck shall accelerate to a speed of 35 mph in not more than 30 seconds.
d. The truck shall attain a top speed of not less than 55 mph.
e. The service brakes shall control and hold the truck on a 30 percent grade. The service brakes shall be tested with the truck heading upgrade and/or heading downgrade. The service brakes shall stop the truck within 35 feet from a speed of 20 mph on a dry, smooth, level surface free from loose material.
f. Parking brake shall be capable of holding the truck in position on a 30 percent grade with truck headed in either direction.
g. The aerial ladder shall be capable of being elevated out of bed, rotated 90°, and extended to the full height smoothly and without undue vibration in not more than 60 seconds.
h. The outriggers shall be capable of stabilizing the truck for safe aerial operation.

3.7 General design.

3.7.1 DoT Federal Motor Vehicle Safety Standards. The truck shall comply with all DoT Federal Motor Vehicle Safety Standards in effect on date of manufacture.

3.7.2 Air pollution controls. The truck shall comply with EPA regulations governing Control of Air Pollution from New Motor Vehicles and New Motor Vehicle Engines applicable on the date of manufacture. In addition, trucks with State of California destination requirements shall comply with State of California Regulations governing air pollution control applicable on the date of manufacture.

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3.7.3 Curb weight. Curb weight shall include weight of truck-tractor chassis and cab and trailer; accessories and equipment; body with all specified firefighting equipment; and full complement of fuel, lubricants, and coolant.

* 3.7.4 Gross combination weight. The gross combination weight (gcw) of complete vehicle shall consist of curb weight, operator, four-person crew, tillerman (weight of personnel computed at 175 pound (lb) each), and payload to provide not less than 60,000 gcw.

3.7.5 Ratings. Vehicle rating shall be the manufacturer's published rating. Component and vehicular ratings shall not be raised to meet the requirements of this specification. When published ratings are not available, verification of ratings must be submitted to the contracting officer. Suspension, axle, and tire capacities shall be not less than 16,000 lb on front axle, 28,000 lb on drive axle, and 16,000 lb on trailer tiller axle which provides for a minimum of 60,000 gcw.

3.7.6 Sound level. The exterior sound level produced by the truck shall not exceed the noise limits established by the EPA. The interior sound level shall be in accordance with DoT Federal Motor Carrier Safety Regulations, section 393.94.

* 3.7.7 Rustproofing. The vehicle shall be rustproofed in accordance with FED-STD-297.

3.8 Tractor components.

3.8.1 Frame. The tractor frame shall provide a rigid platform and shall be bolted construction. The frame material shall be either ASTM A36/A36M steel, or high strength heat-treated alloy steel of sufficient width, depth, and thickness that will provide the required resistance bending moment to support the intended load. When operating the fully equipped and loaded truck under conditions specified herein, the frame shall show no evidence of permanent deformation or distortion.

3.8.2 Fifth wheel. A fifth wheel of sturdy construction shall be mounted at the rear of the tractor frame. The construction shall allow an oscillating motion between tractor and trailer during operation on uneven road conditions.

3.8.3 Engine. The truck engine shall be a diesel engine having horsepower (hp), torque, and speed characteristics to meet satisfactorily all mobile performance requirements specified herein. The diesel engine shall be a liquid cooled, compression ignition, two-stroke or four-stroke cycle engine, with not less than six cylinders. The truck shall meet the mobile performance requirements specified herein utilizing diesel fuel. The engine shall be fitted with the engine manufacturer's standard or recommended fuel filter system. The engine manufacturer's standard "engine start-up" procedure (engine tune-up) shall be applied before delivery of the completed fire truck.

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3.8.3.1 Heavy-duty cooling system. A heavy-duty cooling system shall be provided. A heavy-duty cooling system shall be of the pressurized type of sufficient capacity to maintain a cooling liquid temperature of not less than 140°F, and to provide proper cooling without overheating or loss of coolant when operated in an ambient temperature range of -25°F to +125°F during operating conditions of either stationary or mobile performance as specified. The engine coolant system shall be fitted with silicone rubber hose.

3.8.3.2 Coolant recovery system and indicators. The coolant system shall include a de-aeration system, a surge tank, or a coolant recovery reservoir of not less than 2 quart capacity. In addition, a low coolant level or high coolant temperature alarm buzzer and red indicator warning lights shall be provided on the cab instrument panel.

3.8.3.3 Lubrication system. Engine shall be provided with full forced-feed lubrication. The system shall include a magnetic drain plug in the crankcase and a conveniently located oil fill pipe of sufficient size to permit easy filling when tractor is at a standstill with engine running at full speed. Full flow type oil filter shall be furnished.

3.8.3.4 Governor. An engine governor shall be provided and set to limit engine speed to the maximum speed recommended by the engine manufacturer.

* 3.8.3.5 Engine hour meter. An engine hour meter having a totalizing mechanism of not less than 9,999 hours shall be furnished to register accurately the number of hours of engine operating time. Meter shall be of strong rugged construction to insure continuous trouble-free performance under the most extreme equipment operating conditions indicated in this specification, and shall be so mounted in the truck cab so that it can be conveniently and easily read by personnel servicing the truck. The hour meter shall be the oil pressure type or a combined hour meter with tachometer.

3.8.4 Exhaust system. Exhaust system shall conform to DoT Federal Motor Carrier Safety Regulations, section 393.83. Where necessary, exhaust pipe shall be insulated and shielded. Muffler and tailpipe shall be aluminized steel.

3.8.5 Fuel system. One fuel tank having not less than 50 gallon capacity shall be furnished. The tank shall be of welded construction, shall be baffled, and shall be equipped with a readily removable strainer. The opening for filling the fuel tank shall be so located that when filling the tank, fuel will not spill on the equipment or exhaust pipe. Filler openings must be of sufficient size to permit filling of tank while engine is in operation. The tank and lines shall be amply protected against damage and excessive heat. The fuel tank filler cap shall be the captive chained type. The fuel tank shall have a drainage outlet at the lowest part of the bottom of the tank. The fuel tank shall not extend any lower than the chassis/carrier axle components. The fuel system shall be provided with a fuel filter installed either at the fuel tank or in the fuel line.

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* 3.8.6 Electrical system. The truck shall be provided with a complete 12 volt (V) negative ground starting and lighting system. Electrical equipment and systems, including all lights, reflectors, and wiring shall conform to DoT Federal Motor Carrier Safety Regulations, sections 393.12, 393.13, 393.14, 393.19, 393.20, and 393.22 through 393.33 except as specified herein. Directional signal lamps shall be in accordance with SAE J588 and shall be arrow shaped, amber in color front and rear. The operating unit shall be in accordance with SAE J589, class A. Hazard warning signal switch, in accordance with SAE J910, which flashes all turn signal lamps simultaneously, shall be furnished. Wiring shall be weatherproof, resistant to moisture and fungus, and color coded or numbered for ready identification. Wiring identification shall be consistent throughout the truck. The truck chassis electrical circuits and the truck body electrical circuits shall be separate circuits. The trailer body electrical circuits, excluding truck tractor chassis electrical circuits, shall be provided with overload protective devices of the automatic or manual reset, overload-circuit-breaker type. Two spare single pole, 15 ampere (A) breakers shall be furnished for future radio use. All lamps installed in rear of trailer body shall be flush mounted and so located to not interfere with firefighters climbing up and down rear steps to tiller platform. All lights specified herein shall have chromium plated metal housings. The rear marker lamps in accordance with DoT Federal Motor Carrier Safety Regulations 393.12b shall be located as follows:

a. The three identification lamps located on centerline of body shall be mounted at the highest location on rear of body as practicable.
b. Side rear marker lights shall be located on rear edge of body on horizontal centerline of the three identification lamps. Additional side rear marker lamps shall be mounted in recessed areas, one in each rear corner of rear bumper/step.

3.8.6.1 Trailer lighting and wiring cables. A trailer lighting cable conforming to SAE J1067, with a connector conforming to SAE J560 shall be furnished with its conductors connected and color coded as specified therein. The cable shall be of the coiled type and shall be not less than 110 inches in length. The front of the semitrailer shall be equipped with a receptacle conforming to SAE J560 with its conductors connected and color coded as specified therein. Additional cable(s), conforming to SAE J1067 shall be furnished to provide electrical connection between truck tractor and semitrailer for the following:

a. Emergency lighting.
b. Tillerman to driver signal.
c. Service lighting system.
d. Electrical power to ladder control panel.
e. Compartment lighting.
f. Outrigger warning light on cab dash board.
g. Engine start control.
h. Back-up alarm.

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The cable(s) shall be not less than 110 inches in length and shall be of the coiled types. The connector(s) on end of cable(s) shall not be compatible with the SAE J560 connector to prevent incorrect connecting into the DoT lighting system. When more than one additional cable is furnished to accommodate the required circuits, the connectors shall not be compatible to prevent incorrect circuit connections. The front of the semitrailer shall be equipped with receptacle(s), compartible with respective cable end connector(s), with its conductors connected to trailer circuits.

3.8.6.2 Starting system. The manufacturer's standard starting system shall be provided, complete with an ether or glow plug starting system, and keyless switch with pilot light. When an ether aid system is furnished, it shall be of the measured shot type, controlled from the driver's compartment, with a reservoir of not less than 12 fluid ounces, and inoperative with the engine warm.

3.8.6.3 Alternator. An alternator shall be furnished, complete with rectifier and regulator. Alternator shall be rated for not less than 145A.

3.8.6.4 Battery system. A 12V battery system, with a minimum of two batteries, shall be furnished. Each battery in the system shall be 12V and of the sealed cell (maintenance free) type. Each battery shall be rated in terms of cold cranking capacity of 0°F in accordance with SAE J537. The battery system shall not have less than the engine manufacturer's recommendation of cold cranking ampere required at 0°F for the engine furnished. The batteries shall be installed in "banks" as required. Battery compartments shall be provided in the canopy cab area, as close as possible to the engine in order to reduce battery cable length for connection. Access shall be provided for ease in service and removal. The battery cables shall be protected by use of grommets where the cables pass through metal bulkheads and shall be covered by a protection boot where the cables are attached under vehicle frame to the electrical switches or connectors. The battery compartments shall be ventilated and treated against acid fumes and spillage. The batteries shall be provided with a polorized receptacle(s) for station charging. The receptacle shall be located near the batteries at the driver's entrance to the cab. The charging receptacle shall be suitably identified. The receptacle shall be of the recessed male type with a spring loaded cover and provided with a matching female plug with protected prongs. Contractors shall also provide the female plug for the receptacle. A carrying handle for the batteries shall be provided. Removable battery compartment cover(s) shall be fitted with lifting handles. Finger lift openings are not acceptable.

3.8.6.5 Service and emergency lighting systems. In addition to the standard DoT running lights specified in 3.8.6, additional service and emergency lighting shall be furnished as specified herein.

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* 3.8.6.5.1 Service lighting system. The truck shall be equippped with a service lighting system as follows:

a. Cab dome lights shall be furnished in the driver's compartment and in the rear canopy area.
b. Two automotive type, 6-inch sealed beam, roof type spotlights shall be mounted to the roof of the cab. One spotlight shall be located near the left windshield post and one shall be located near the right windshield post. Controls shall be extended inside the cab for ON-OFF control and movement of lamp housing through a horizontal included angle of at least 270°, and a vertical angle of at least 90° above and 35° below horizontal.
c. Two backup lights, flush mounted at rear of body. The lights shall be automatically activated when vehicle is in reverse gear operation. The lights shall be not less than 6 inches in diameter.
d. Suitable lighting at the base of the aerial ladder to illuminate approaches to the ladder control station and also two spotlights to illuminate the ladder in any position of operation.
e. Suitable lighting shall be provided for all instruments, gages, ladder panel controls and outrigger controls.
f. Compartments specified in 3.10.14 shall have one or more lights for illumination. Each light shall be similar or equal to 2-1/2 inch model manufactured by the Griffin Lamp Company. A master control switch shall be mounted in cab instrument panel with flashing pilot light to indicate "ON" position and identified as to function.
g. Two adjustable pedestal type chrome-plated spotlights for area lighting shall be mounted on the rear of the trailer. Switch to be on the light and shall be weather resistant.
h. Four Collins FX-12, or equal, 12V spotlights shall be mounted, two on the base section, and two on the top of the ladder section. Each light shall have an on/off switch on the light. Lights shall be mounted below handrail height so as not to increase the overall height of the unit. A master switch for these spotlights shall be provided on the turntable console.

* 3.8.6.5.2 Emergency lighting system. The truck shall be equipped with an emergency lighting system as follows:

a. Two revolving 12V warning lights shall be provided and mounted with reinforcement at point of attachment. Each warning light shall have a clear dome, shall have four sealed beam lamps each having a minimum of 35,000 candlepower (cp) and a minimum flash rate exceeding 80 flashes per minute. Two of the lamps shall be clear and two shall be red. The lamps shall be installed in alternating colors. The minimum warning light dimensions shall be: 9 inches high, 9-inch diameter, 5-inch skirt (base). The warning lights shall be mounted on the cab roof, one on each side, located approximately 6 inches in board from the edge of the cab roof and

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approximately midway between the windshield and the forward edge of the canopy roof.

b. Two revolving warning beacon lights similar and equal to KD brand lamps, style KD886, having a rotating flash rate exceeding the 80 flashes per minute. The beacons shall be mounted on top of rear body cabinets, one on each side, in a location which does not interfere with firefighters climbing to and from ladder base.

c. Two red flashing warning lights shall be mounted on the front of the truck, installed on a horizontal plane through the centerline of the turn signals. The warning lights shall be approximately 7 inches in diameter similar and equal to Weldon model 1020 having a minimum 200 cp. A heavy-duty flasher unit shall be furnished which provides for a combined alternating flash rate of not less than 80 flashes per minute.

d. Side marker lamp. A series of red side marker lamps, mounted to bottom edge of compartments (left and right), shall be furnished. One or more shall be furnished behind the rear wheel housing. Not less than five shall be furnished forward of rear wheel housings spaced on not greater than 30-inch centers. The lamps shall be similar or equal to Grote model 46352 with red lens.

e. Two 6-inch sealed beam red flashing warning lights shall be located at the front of the semitrailer, one on each side with the light directed to the side mounted on.

f. A single control switch to operate the front and rear revolving warning lights, front flashing lights and red side marker lamps, shall be mounted in the instrument panel, accessible to the driver. A red pilot light, indicating the "ON" position, located in the instrument panel, near the control, shall be provided. The switch shall be identified as to function.

* 3.8.6.6 Siren. An electrical motor driven siren equal to Federal Q2b type, pedestal mounted on the driver's side of the bumper extension, shall be furnished. The siren shall have floor mounted activator devices (driver and passenger). A siren brake control shall be mounted at the drivers' side of the instrument/dash panel, identified, and in a location readily visible to the driver.

* 3.8.6.7 Horn. An electric horn of manufacturer's standard type shall be furnished. Also, two chrome plated air horns of the studder tone type shall be provided on the cab roof, one horn on each side of the roof. The air horns shall be controlled from one floor mounted, foot operated control, located on the fire officer's side of the cab and a lanyard (pull type) control from the overhead center of the cab.

* 3.8.7 Transmission. An automatic or semiautomatic transmission shall be provided. The transmission shall include a hydraulic torque converter and not less than five forward gear ratios. The gear ratios shall be of the wide ratio type which provides a first gear ratio of not less 7.5. The shift

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control shall provide for driver selection of individual gears or ranges of gears. The transmission shall be provided with SAE J704 power takeoff opening(s). Net torque capacity of the transmission shall be not less than 480 pound-feet and a net input power rating of not less than 250 hp. The transmission shall have a torque converter lockup in the two highest forward gears. Torque converter multiplication factor shall not be used to establish first gear ratio of not less than 7.5.

3.8.7.1 Transmission capacity. The torque input capacity of the converter transmission assembly shall be in excess of the net torque output of the engine, and the maximum speed output of the engine shall not exceed the rated input speed of the transmission. The converter with transmission design shall acknowledge net flywheel hp.

3.8.7.2 Torque converter. The torque converter shall be heavy-duty, having a multiplication factor of not less than 2 to 1 at stall. The torque converter shall be equipped with a heat exchanger. The torque converter transmission combination shall be provided with an oil filter on the low pressure side of the lubricant fluid level.

3.8.7.3 Converter gage. A torque converter temperature gage, suitably mounted on the instrument panel, complete with flashing warning light that will be clearly visible to the operator when the temperature of the converter fluid exceeds the unsafe temperature, shall be provided. The signal light shall be suitably identified with a permanent metal plate. An additional audible device shall be provided which will actuate the horn when the converter fluid temperature exceeds 275°F.

3.8.8 Drive line components. Drive line components shall be adequate to transmit the maximum delivered torque of the engine, as developed through the maximum gear train reduction.

3.8.9 Axles. Axle ratings shall be at least equal to the load imposed on each axle, measured at the ground, with the tractor-trailer loaded as specified in 3.7.4.

3.8.10 Suspension. Tractor shall be equipped with suspension system with components having a rated capacity at least equal to the load imposed on each member, measured at the ground, with the tractor-trailer loaded as specified in 3.7.4. Hydrualic double-acting shock absorbers shall be provided at front and rear wheels.

* 3.8.11 Back-up warning device. The truck shall be provided with an automatically activiated, audible, pulsating, signaling warning device, mechanical or electrical, to caution personnel when truck is in reverse gear operation. The back-up alarm shall have a noise level of at least 100 decibels at a distance of 10 feet.

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* 3.8.12 Brakes. Brakes shall conform to DoT Federal Motor Carrier Safety Regulations, sections 393.40 through 393.43, 393.45 through 393.52, and as specified in the current FMVSS-121 Standards.

3.8.12.1 Service brakes. The service brakes shall be of the air application type. Air filters shall prevent brake system contamination. Brake operation shall not be subject to excessive fading.

* 3.8.12.1.1 Brake system air compressor. The capacity of the air compressor shall be sufficient for the intended service but shall be not less than 15.5 cubic feet per minute. The air compressor shall be equipped with a governor and the compressor intake shall be properly connected to an air cleaner. An air drier, Bendix Model AD-4 or equal, shall be furnished and installed to remove moisture from the air. The moisture removed from the air by the air drying system shall be disposed of automatically and the moisture collector shall be winterized to prevent freezing.

* 3.8.12.1.2 Brake system air storage reservoirs. The air storage reservoir shall have two air gages, with indicator lights and buzzer alarm located inside the operator's cab. The system shall be of the quick-release buildup valve with the air compressor connected to a small reservoir which is pressure-relieved into the larger reservoirs when the minimum pressure is reached for optimum operation of the brake system. The time required to reach this pressure from zero pressure at 50 percent engine governed speed shall not exceed 30 seconds. If other than brake air supply requirements are to be supplied from the reservoirs, a priority system shall be installed to assure that an adequate supply of air is always available to operate the brake system. All reservoirs shall provide a minimum of 4500 cubic inches and have manually operated, conveniently located, water drain valves.

* 3.8.12.2 Parking brakes. Parking brakes, with independent operation and control from the air service brake system, shall be provided. The parking brakes shall be spring type on all rear wheels and have an indicator light and control valve located within reach of the driver. The parking brake shall be capable of meeting the performance requirements specified in 3.6.

* 3.8.12.3 Trailer brake control system. In addition to the components specified in 3.8.12.1, a trailer brake control system shall be furnished and shall include:

a. Identification of emergency and service lines.
b. Coincident control of trailer brakes with prime mover foot control.
c. Independent hand control for trailer brakes.
d. Prime mover protection valve with dash control and automatic break-away feature.
e. Trailer stoplight control operative with foot brake and with hand control for trailer brakes.

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f. Two connecting air hoses, not less than 110 inches in length, equipped with coiled spring hose guards.
g. Spring supports on cab or on a pogo stick type hose tender.

* 3.8.13 Tractor wheels, rims, tires, and tubes. Tractor shall have single front and dual rear wheels. Rim and tire ratings shall conform to TRA recommendations. Tires shall be tube or tubeless type with highway tread. Tires shall be steel belted radials. All tires shall be of the same size and ply rating. The rated individual tire capacity at 50 mph highway conditions shall be mot less than the maximum wheel loading imposed with truck loaded as cited in 3.7.4. Each wheel assembly shall be balanced individually. Wheel and tire assemblies shall be dynamically balanced at simulated speeds up to 60 mph. Disc type wheels shall be provided.

3.8.13.1 Spare wheel. A spare wheel or rim with inflated spare tire shall be furnished for the tractor, and stowed in a protected location on the truck for shipment.

3.8.14 Steering mechanism. The truck shall be equipped with a power assist steering mechanism of hydraulic type. The mechanism shall permit manual steering in the event of failure of the hydraulic system, and shall be properly effective at all truck speeds. The system shall be furnished complete with all necessary piping, valves, and accessories. The steering mechanism shall provide for controlling the direction of the fully loaded truck under all operating conditions. The cramping angle shall be not less than 30° right and left.

* 3.8.15 Cab. The cab shall be five-person fixed canopy type (non-tilt). The canopy portion of the cab shall allow for readily accessible engine maintenance and removal. The engine hood (cover) in the canopy cab section shall be insulated against heat and noise, but shall allow for adequate venting away from seated personnel.

3.8.15.1 Cab construction. The cab shall be of the all aluminum reinforced, welded construction. The cab interior width dimension shall be not less than 88 inches. With the driver's seat in the rearmost position, cab width shall be measured at the floor on a transverse line through a vertical plane, 6 inches forward of the driver's seat cushion. Aluminum sheet metal used in construction of the cab shall be at least 0.125-inch thick. The cab shall provide unobstructed seating and head room for a minimum of five persons, fully clothed in firefighting gear. In the forward section of the cab, seats shall be provided for two passengers and driver, and in the rear, under the canopy section, two jump seats shall be furnished. The two passenger seats in the forward cab will be jump seats designed to accommodate an air pack as described in 3.8.15.2. The forward enclosed section of the canopy type cab shall be provided with a door on each side of the fire truck. The cab doors shall be ample size for easy access to the forward cab and shall be equipped with roll-down windows. The doors shall be provided with locks

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and with at least the front curbside door equipped with an external key-operated lock. Grab handles shall be provided on each side of the fire truck at each cab door and inside the cab at each end of the instrument panel. Additionally, two grab handles shall be provided at each canopy entrance of each side of the truck. The outside grab handles shall be of a nontwist design, polished stainless steel, and shall be not less than 1-1/4 inches in diameter. A space shall be provided between hand rail and cab body for grasping with heavy gloved hand. A step shall be provided at each cab door and at each canopy entrance. Each step shall have an aluminum, nonslip tread surface. Each step shall support, without permanent deformation, the weight of a 250-lb firefighter in full firefighting gear. Each step shall be of maximum width and depth to permit safe entry and exit for firefighters wearing firefighting foot gear. Safety glass shall be provided throughout the cab. Windows between the forward cab and canopy type shall be sliding glass with finger pulls and equipped with a latching device. The channel for the sliding window shall be lined to reduce rattles. A dome light shall be provided in the forward cab, two step lights shall be provided, and two dome/map lights shall be provided in the rear canopy section; the forward dome light and two step lights shall be controlled by door and manual switches. Rubber boots shall be provided for all control floor board openings. Dual sun visors shall be provided. Cab floors and walkways shall be of heavy gage aluminum diamond plate. A full circular, wheel well liner shall be furnished to protect against dirt and water entry. The forward cab shall meet the interior noise criteria of DoT Federal Motor Carrier Safety Regulations, section 393.94.

* 3.8.15.2 Seats. All seats shall be upholstered over deep air foam cushions with first grade black vinyl covering, suitable for fire service. All seats shall be equipped with an adjustable headrest restraint. The driver's seat shall be a heavy-duty, four way adjustable air ride bucket-type seat similar or equal to Anchorlok brand. The fire officer/passengers and the two rear facing under canopy seats (jump seats) shall by Ziamatic Quick-seat TM or equal. Each jump seat shall have a padded lower cushion of suitable width and depth to provide ample seating for firefighters wearing turn out gear and back strapped air packs. The back of each jump seat shall have an air breathing apparatus walk away bracket, Ziamatic Model U45F or equal. The bracket shall provide for mounting of the air breathing apparatus in upright position. A series of predrilled holes shall be provided in jump seat backs for vertical adjustment of walk away brackets. Each jump seat back shall be provided with padded flanges to each side and forward of the breathing apparatus. Seat belts shall be provided for the driver and passengers jump seats of sufficient length to accommodate firefighters wearing turn out gear. The rearward facing jump seats shall provide additional storage compartment(s) if available in manufacturer's basic design. The right rear facing jump seat shall have at least 4 inch spacing between seat and cab exterior wall and the left rear facing seat, (behind driver) shall have at least 4 inch spacing between seat and engine enclosure. A space of at least 18 inches between bottom of jump seat and canopy decking shall be provided.

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3.8.15.3 Controls and operating mechanisms. All controls and operating mechanism shall be located for left-hand drive. Controls shall be complete and conveniently operable by the driver. Lever controls shall be designed and located to permit easy entrance and exit of operator to and from driver's compartment. Instruments and controls shall be identified as to their function and installed in a manner to facilitate removal and servicing.

* 3.8.15.4 Cab accessories and equipment. Equipment shall be complete with all accessories furnished as standard equipment by the manufacturer. The following minimum equipment shall be furnished and suitably identified:

a. Keyless starting switch with pilot light.
b. Ammeter gage or voltmeter or charging indicator.
c. Fuel gage.
d. Oil pressure gage.
e. Engine temperature gage.
f. Speedometer with recording odometer.
g. Tachometer.
h. Dual sunvisors.
i. Hour meter.
j. Torque converter temperature gage.
k. Service brake air pressure gage.
l. Brake quick build-up gage.

* 3.8.15.5 Heater and defroster. Enclosed portion of the cab shall be equipped with a hot water type heater having fresh air intakes, discharge outlets to floor, and windshield defroster louvers. Heater shall be equipped with a 12V, electric-motor-driven, two-speed blower controlled by a dash mounted, three-position switch. The location of the heater shall not restrict or interfere with personnel when entering or exiting the cab or while seated.

3.8.15.5.1 Defrosting, anti-icing, and ventilating. A sufficient defrosting, anti-icing, and ventilation system shall be provided in the cab. This system shall have the capacity to effectively prevent formation of frost in the interior surface of all transparent areas of the cab and to prevent the formation of ice on the exterior of the vehicle windshield. The system shall also maintain clear vision over at least 75 percent of the area of all windows of the cab interior.

3.8.15.5.2 Controls. Variable controls shall be provided on air blowers and at least high output setting shall be furnished on the cab heating system controls. Adequate operating instructions shall be provided at heater system control point. Cab heater and defroster system control shall be mounted in the instrument panel.

3.8.15.6 Outside rearview mirrors. Outside rearview mirrors shall be mounted on each side of the cab. Each mirror shall have at least three supporting arms or heavy-duty spring loaded swing away ratchet type side

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mounting brackets. Combination (flat and convex) mirrors shall be furnished. Each combination mirror shall have at least 50 square inches of flat reflective area and an adjustable convex surface having at least 20 square inches of reflective area.

* 3.8.15.7 Windshield wipers. Heavy-duty electric operated windshield wipers shall be mounted to facilitate clear vision. The wipers shall be of such capacity and coverage as to readily cleanse the windshield of rain, snow, and ice when used in conjunction with the defrosting system or windshield washer mechanism. The windshield wipers shall meet the requirements specified in SAE J198. Wiper assemblies shall include features to permit speed control and return of the wiper blades to a retracted position out of the line of vision when not in use.

3.8.15.8 Windshield washers. Power operated windshield washers, equipped with a minimum of 1 gallon capacity plastic container for cleaning fluid, shall be furnished.

3.8.16 Running boards and kickplates. Running boards, fabricated from heavy-gage aluminum diamondette, shall be furnished on each side of the tractor at the canopy entrances. Aluminum kickplates shall be furnished at each cab and canopy entrance.

* 3.8.17 Front bumper and towing devices. A heavy duty, full width, front bumper, and two tow hooks shall be provided at the front of the tractor. The front bumper shall be polished stainless steel. The front bumper shall project out at least 14 inches in front of the cab to provide a step at the front of the truck. A grab rail, not less than 12 inches long, shall be provided on the cab front below the windshield, to facilitate personnel stepping up and onto the extended bumper. The two front tow hooks shall be mounted to the frame or rigid members which are attached to the frame. When access to the two front tow hooks are furnished through the front cab panel, access holes with hinged cover plates shall be provided.

3.8.18 Power transfer controls. Controls shall be installed at the driver's position for transferring the power from road drive to the aerial ladder mechanism.

3.9 Hydraulic system.

3.9.1 Hydraulic system pump. The pump, which supplies the liquid for all hydraulic operations, shall be the manufacturer's standard type driven by power takeoff from the truck engine. The pump operating lever shall be located in the driver's compartment, shall be provided with a lock mechanism to prevent accidental engagement or disengagement, and shall be suitably marked for identification and operating procedures. The pumping system shall include an adjustable pressure relief valve or unloading valve with the range of adjustment which allows raising or lowering of the aerial ladder with any working or test loads specified herein.

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3.9.2 Hydraulic fluid reservoir. A hydraulic fluid reservoir shall be provided and shall have sufficient capacity to hold the required volume of hydraulic fluid with not less than 10 percent for expansion space. The reservoir shall be provided with a filler opening and removable oil strainer and a vented filler cap, accessibly located to facilitate filling the tank from a 1-gallon container using a funnel. The vented filler cap shall have means to preclude fluid from splashing out of the tank during movement of vehicle. The filler cap shall be attached to the tank by a corrosion-resistant chain to prevent loss. Provisions shall be made for inspection and cleaning of interior and draining of the reservoir. The reservoir shall have a photo-sensitive corrosion-resistant plate indicating the type of hydraulic fluid required and the maximum level in the reservoir with the aerial ladder in travel position and with the hydraulic lines and components full of fluid. A sight level or dipstick gage shall be furnished to indicate level of fluid.

3.9.3 Oil filter and strainers. A full-flow type 10-micron discharge oil filter in the return line to the reservoir shall be provided. A 100-mesh suction line strainer in the hydraulic reservoir shall be provided. Filter and suction strainer shall be easily accessible for periodic cleaning, or for replacement of the filter cartridge. Filter and strainer shall have sufficient capacity to allow the maximum rated pump discharge.

3.9.4 Hydraulic lines and fittings. Hydraulic lines shall be located to facilitate maximum ease of removal and installation and visual inspection for failures and maximum protection from damage during operation of the aerial ladder. Hydraulic lines and fitting shall be air and liquid tight and shall be installed to preclude vibration and chafing. Provisions shall be made for bleeding air from the hydraulic system and completely draining all hydraulic fluid lines. Hose assemblies and all seamless steel hydraulic lines shall be adequate to provide the rate of flow, temperature, and pressures required to operate the system smoothly and efficiently. Galvanized pipe and copper tubing shall not be furnished.

3.9.5 Hydraulic gage. An oil pressure gage to indicate hydraulic pressure shall be provided at the ladder operator's position.

3.10 Trailer components.

3.10.1 Frame. Trailer frame construction shall have the structural capacity to withstand the flexural forces during operation of the aerial ladder, as well as the mounting of all required firefighting components. Main frame member shall be adequately braced with a sufficient number of crossmembers to present a rigid construction with sufficient strength for loads imposed.

3.10.2 Axle. Trailer shall be provided with a steering rear axle. The axle rated capacity shall be at least equal to the load imposed, measured at the ground, with the tractor-trailer loaded to gcw.

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3.10.3 Suspension. Trailer shall be equipped with a suspension system with components having a rated capacity at least equal to the load imposed, measured at the ground, with the tractor-trailer loaded to GCW.

3.10.4 Rear steering. Steering gear shall be of the design that eliminates transmission of road shock to the steering wheel and provides easy and safe steering. The large diameter steering wheel shall be located at the correct position and angle for easy steering. A steering gear lock shall be provided which will lock the tiller steering in the center position for transporting the tractor-trailer unit when a tiller driver is not required or available. The lock shall be manually operated and shall not interfere with tiller operation when in the unlocked position.

* 3.10.5 Trailer service brakes. Service brakes shall be of the full air type, and shall conform to DoT Federal Motor Carrier Safety Regulations 393.40, 393.42, 494.43, and 393.45 through 393.47; except in lieu of conformance to the brake performance requirements referenced therein (393.52), the vehicle shall conform to 3.8.12.3. The braking system shall include standard breakaway features, relay emergency valve, air reservoirs, slack adjusters, piping, hose connections, and all other components required for a complete air brake system. The air hose location shall comply with SAE J702. The braking system shall be installed in a manner which provides road clearance for travel over uneven terrain and protection against damage caused by objects striking the components. No part of the braking system shall extend below the bottom of the wheel rims. The air reservoirs shall be of sufficient capacity to supply air for the air brakes and the complete pneumatic control system of the trailer, with an adequate reserve capacity.

* 3.10.5.1 Trailer parking brakes. The trailer shall be equipped with spring applied, parking brakes on each wheel which shall hold the trailer with rated payload, on a 10 percent grade despite the depletion of the compressed air supply. The parking brake shall be automatically applied upon disconnection of the emergency air line, and under emergency braking conditions, and shall remain applied without any further application.

3.10.6 Trailer wheels, rims, tires, and tubes. Trailer shall have single rear wheels. Rim and tire ratings shall conform to TRA recommendations. Tires shall be tube or tubeless type with highway tread. Tires shall be steel belted radials. All tires shall be of the same size and ply rating. The rated individual tire capacity at 50 mph highway conditions shall be not less than the maximum wheel loading imposed with tractor trailer loaded as cited in 3.7.4. Wheel and tire assemblies shall be dynamically balanced at simulated speeds up to 60 mph. Disc type wheels shall be provided.

* 3.10.7 Rear platform and running boards. Rear platform and running boards, fabricated from heavy-gage aluminum or steel "diamondette", shall be furnished. Aluminum or steel kick plates shall be furnished at forward edge of rear platform. The rear platform shall have a load carrying capacity of not less than 700 lb. The rear platform shall be full width of body and not less than 15 inches from rear edge to rear of body. The corners of the platform shall be rounded or formed at a 45° angle with recess for mounting of

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rear side lower marker lamps. The rear platform shall be not greater than 22 inches from ground level with tractor trailer loaded to gross combination weight (GCW).

3.10.8 Tillerman's cab. A fully enclosed tillerman's cab shall be provided. A fixed type tillerman's seat, with back, shall be provided and shall be durably upholstered. The windshield shall be of safety glass, and sloped to eliminate glare. A power operated constant speed windshield wiper shall be provided. A suitable signal shall be provided for the tillerman to the tractor driver. The signal shall be both visual and audible to the driver in the tractor cab with signal controls at the tiller seat.

3.10.9 Tillerman's platform. A tillerman's platform shall be provided, and a large step on each side of the ladder bank shall be provided to facilitate mounting. The platform and steps shall be of heavy-gage nonskid steel or heavy-duty aluminum diamond plate.

* 3.10.10 Aerial ladder. The aerial ladder should consist of a minimum of four sections and extend to a nominal working height of not less than 100 feet above the ground. Ladder design should incorporate a minimum safety factor of three to one, based on the live load. A one and one-half to one stability factor (in compliance with the OSHA and ANSI) is required. Ladder construction should be of welded, steel tubing with a minimum of 46,000 pound-force per square inch (psi) high-strength. Each section of the ladder shall be trussed diagonally, vertically and horizontally, and should be reinforced at critical points to provide extra rigidity. All ladder rungs are to be round and welded to each section using "K" bracing. All rungs are to be covered with serrated replaceable heavy-duty rubber sheaths, which are to be glued and clamped securely to the rungs. Inside width dimensions of the four ladder sections shall permit the passing of personnel on the ladder and be no less than 36 inches wide at the base section and no less than 20 inches wide on the fly section. The height of the handrails above the rungs shall be no less than 26 inches on the base section 23 inches on the inner section, 19 inches on the outer section and 15 inches on the fly section. Folding steps will be provided for the operator's footing while at the tip of the fly section. A ladder rest shall be provided to support the ladder while in the travel position. Mounting brackets shall be provided on the inside of the fly section for an axe and a 10-foot pike pole. A telescopic water pipe system shall be provided on the ladder and a monitor with nozzle equal to the Akron model 3478. The piping sizes shall be a minimum of 4.5 inch outside diameter (od) in the base section, 4 inch od in the inner mid-section, 3.5 inch od in the outer mid-section and 3 inch od in the fly section. A pipe will be provided to connect the water supply on one end and to a water swivel coupling that will allow the ladder to rotate 360° continuously while pumping water. A pipe will be routed through the hydraulic and electric swivels to the heel pin swivel to allow water to flow to the ladder pipe while elevating the ladder from - 6° to +75°.

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* 3.10.10.1 Raising device. A minimum of two double-acting lift cylinders shall provide elevation from at least - 6° to +75° above horizontal. The cylinders shall be equipped with holding valves that will prevent the unit from dropping, even if a charged hydraulic line should fail anywhere in the system. A fully hydraulically powered extension and retraction system using two hydraulic cylinders, with each cylinder being able to operate the ladder should one of the cylinders fail, shall be provided. The aerial ladder capacity, in an unsupported position, based on 360° rotation, and full extension from -6° to +75°, shall be a minimum of 400 lbs at the tip. There shall be two gages at the control console that show the system hydraulic pressure and the ladder load. An alarm shall be provided that will have an audible and visual warning that will activate when the ladder load capacity is exceeded. The aerial water system shall be able to pump 1000 gpm while fully extended and 90° to each side of center with 400 lb load at the tip. Adequate hand clearance shall be provided between the sides of the ladder and moving cylinder(s) or other moving parts, or suitable hand guards shall be provided to prevent injury. A calibrated safety valve in the return oil passage of the cylinder(s) shall automatically control the lowering speed of the ladder. The ladder control levers for all ladder operations shall be grouped in a operator's control pedestal located on the platform. The aerial ladder shall be capable of being elevated from the ladder rest, rotated 90°, and extended to the full height smoothly and without undue vibration in not more than 60 seconds.

* 3.10.10.2 Platform and turntable. A sturdy operating platform, built of heavy-duty aluminum diamond plate shall be provided and allow for ample unobstructed area for all rescue and service operations. The turntable shall provide 360° rotation in either direction by hydraulic power, and shall provide maximum ladder operating stability with minimum rotation resistance. Steps and handholds or rails shall be provided to reach the turntable from either side. The steps shall be illuminated. The turntable platform shall be provided with a safety railing on each side of the platform of 1-1/4 inch polished stainless steel tubing at least 36 inches high and shall be of the nontwist type. Access/clearance, as specified in 3.16, shall be provided for the turntable mounting bolts so that correct torque can be checked in accordance with procedures specified in NFPA 1904. The turntable treadplate shall have downward facing of at least 2 inches. Anti-skid strips shall be applied at the base of the ladder on the treadplate. The turntable shall be well lighted by lights activated from the aerial master switch.

3.10.10.3 Aerial ladder control, locking, and safety devices. Clearly marked, suitably lighted, and fool proof deadman control, locking, and safety devices shall be provided to perform the following:

a. Immobilize the truck.
b. Transfer power application to the ladder mechanism.
c. Lock in and release the ladder from its bed.
d. Stabilize the truck.
e. Control elevation, rotation, and extension of the ladder.

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f. Indicate the angle of elevation, rotation, and extension of the ladder.
g. Visual indication of the safe load at any angle or extension of the ladder.
h. Start and control the engine from the operator's position.
i. Lock control for ladder rung locks or pawls.
j. Provide auxiliary power (see 3.10.10.10) operation of the ladder in the event of power failure.

3.10.10.4 Immobilizations controls. Controls for immobilizing the vehicle shall be installed at the driver's position. In addition, four-wheel chocks are to provided to be positioned by driver at trailer wheels before aerial ladder operation.

* 3.10.10.5 Ladder bed lock. A lock shall be provided which will prevent action and movement of the aerial ladder in its bed when the vehicle is in motion. The lock release or control shall be located at tillerman's position or at the turntable or may be automatically performed when actuating the controls required. An interlocking system of limit switches shall be provided to prevent the lifting of the aerial ladder from the bed lock position until the outriggers have been deployed to the load supporting position and also to prevent any movement of the ourtriggers while the aerial ladder is raised from the bed lock position.

3.10.10.6 Elevation controls. A control for elevating the ladder out of its bed to the maximum angle of elevation and returning it shall be provided at the operator's position. This control shall be arranged to permit the operator to regulate the speed of elevation within safe limits determined by the manufacturer. A manual lock valve or other positive locking device shall be provided so that the desired angle of elevation can be maintained indefinitely without dependence upon engine power.

* 3.10.10.7 Turntable controls. Controls for rotating the turntable 360° in either direction shall be arranged to permit the operator to regulate the speed of rotation within limits determined by the manufacturer. The turntable shall be fitted with strong friction or other positive locking device so as to retain it in any desired position. Suitable indicators, clearly visible at night, shall be provided to facilitate correct alignment of the turntable with the ladder bed. A back-up manual system shall be provided to operate the rotation of the turntable in the event of a loss of power. The control console shall be well lighted with all control switches clearly identified and easily accessible. The control switches for each ladder section shall allow individual and/or simultaneous operation for extending and retracting the ladder sections. An automatic throttle control shall be provided to increase the engine idle speed to a pre-set revolutions per minute (rpm) when the ladder sections are being extended, retracted, raised or lowered, and return the engine to idle speed when the ladder activating system is in the off position.

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3.10.10.8 Ladder extension controls. Controls shall be installed at the operator's position for extending the fly sections of the aerial ladder to any desired height between the fully retracted and fully extended position; the same control may also be used to retract the sections. Safety provisions shall prevent damage at full retraction or extension. Provisions shall be made so that, in the event of failure of extension mechanism during operation, the descent of the ladder shall be kept at a speed which will prevent damage to the ladder structure or danger to personnel. Stops shall be provided to properly align the sections without damage to the ladder or the cable mechanism when the ladder is retracted.

* 3.10.10.9 Ladder lock controls. A lock that controls ladder rung locks or pawls shall be provided at the base of the ladder adjacent to the operator's position to prevent movement of the sections when the power is shut off or in the event of loss of pressure in the hydraulic circuits. This lock shall be arranged for use when aligning rungs. The control for the pawls shall be such that the on and off position may be clearly determined by the operator without it being necessary to see the pawls. To prevent damage to the ladder due to application of power when the pawls are locked, it is desireable to have an automatic device to prevent retraction of the ladder sections until the pawls have been released. The above locking can be accomplished manually and/or hydraulically.

3.10.10.10 Auxiliary power. Provision shall be made so that an auxiliary electric motor shall power the hydraulic system for all ladder functions when normal operating power source fails. If required, manual controls shall be provided which will permit releasing of the pawls for retraction of the ladder in the event the auxiliary power system is used.

3.10.10.11 Engine controls. A supplementary engine starting control shall be provided on or adjacent to the operator's control pedestal. If the engine speed is not automatically regulated or predetermined, a suitable engine throttle shall be provided but shall be governed to prevent operation of the ladder at excessive speeds and a tachometer shall be installed.

3.10.10.12 Angle of elevation indicator and load limit. A device, clearly visible from the operator's position, shall be provided to indicate the angle of elevation of the ladder at all times. Either a load indicator or an instruction plate visible from the operator's position shall show the recommended safe load under existing conditions of ladder elevation.

3.10.10.13 Power transfer control. Controls shall be provided at the driver's position to transfer the power from the road drive to the aerial ladder mechanism.

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* 3.10.10.14 Training. The manufacturer of the aerial ladder and related control systems, shall provide a minimum of 24 hours training to final destination fire department personnel in the operation and maintenance of the aerial ladder. Training to take place at site of receiving customer destination.

* 3.10.11 Outrigger(s). Hydraulic outrigger(s) shall be provided and positioned to fully support the frame and turntable structure. Each outrigger extension shall be not less than 145 inches and shall have positive stops to prevent over extension. Separate controls shall be provided for each outrigger to permit side-to-side leveling of the vehicle on inclined surfaces. A positive system for holding the outrigger(s) in the cradle position shall be provided. A readily visible leveling indicator shall be provided and properly illuminated to indicate proper stabilization of the truck to the operator when setting the outrigger(s). Automatic pilot check valves shall be provided for each outrigger to block the hydraulic fluid in the cylinder and to prevent movement of the outrigger(s) in event of damaged hydraulic line. All outrigger controls shall be installed in a readily accessible location which permits operation from ground level. The outrigger(s) shall withdraw to within the overall dimension of the truck. An audible alarm shall be provided on each side of the truck and will sound a warning when the outriggers are being deployed or retracted. A self-activating locking device shall prevent accidental raising of the outriggers while the aerial ladder is elevated from the storage/travel position.

3.10.11.1 Outrigger warning lights. Red indicator/warning lights, suitably identified shall be furnished at the ladder operator's control pedestal to indicate that the outrigger(s) have been extended.

3.10.11.2 Outrigger warning plate. A warning plate shall be provided, installed at ladder operator's position on the turntable, that reads as follows: DO NOT OPERATE AERIAL LADDER UNLESS THE OUTRIGGER(S) AND AXLE LOCKING JACKS ARE DOWN AND IN THEIR LOCK POSITION.

3.10.11.3 Outrigger floodlights. Floodlights shall be provided at each outrigger station to illuminate the outrigger positioning area.

* 3.10.11.4 Outrigger ground pads. Each outrigger ground pad shall have a ground contact surface area of not less than 140 square inches. Ground pads shall be of the hinge or swivel type. If the ground pad swivels in one direction only, it shall be in a plane normal to the axis of the apparatus. Additional foot plates shall be furnished to increase the ground contact area of each outrigger to not less than 320 square inches. Storage space for the foot plates shall be provided on the vehicle.

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3.10.12 Axle jacks or locks. A suitable device shall be installed which will prevent transmission of the load on the aerial ladder to the rear spring of the tractor. The axle jacks or locks shall lock tractor frame to the tractor rear axle below the ladder turntable to make the chassis springs inoperative, and to prevent excessive side tilt/pull when ladder is in use. The jacks or locks shall be of a positive type which will operate on all types of terrain or gradient on which the truck may be positioned.

3.10.13 Ground ladders. Aluminim ground ladders shall conform to provisions of NFPA 1931, article 100. Extension ladders shall be equipped with the manufacturer's latest design safety locks and raising mechanisms. The length of the ladder shall be the measured length when extended and not the sums of the lengths of the sections. The following ladders, in the quantity shown, shall be furnished and installed on the trailer.

a. Two 35 foot, two-piece or three-piece extension ladder.
b. One 30 foot, two-piece extension ladder.
c. Two 28 foot, two-piece extension ladder.
d. One 18 foot, single ladder with folding roof hooks.
e. One 16 foot, single ladder with folding roof hooks.
f. One 14 foot, single ladder with folding roof hooks.
g. One 10 foot, callapsible ladder.

The ladders shall be carried in a double bank arrangement and locked down by quick-acting locking devices.

3.10.13.1 Ground ladder storage. Ground ladders shall be stored in the trailer bed under the aerial ladder. Access to the ground ladders shall be from the rear of the trailer. Teflon or oak stripping shall be used to protect ladder bed runners against continuous rubbing and chafing during removal and installation of ladders. The ladder stowage space shall be boxed in to prevent entry of ground water and road grit from underside of trailer.

3.10.13.2 Pike poles. The following fiberglass type pike poles shall be furnished and stowed in the truck adjacent to or along side of the ground ladders. If tubing is used for storage, the material shall be aluminum.

a. Two - 6 foot
b. Two - 8 foot
c. Two - 12 foot
d. Two - 16 foot

In addition to the pike poles specified in items "a." thru "d.", the basic accessory equipment listed in paragraph 10-4.1 of NFPA 1901 shall be provided.

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3.10.14 Equipment compartments. Trailer shall be provide with side equipment compartments as specified herein. Equipment compartment arrangement shall include at least the following:

a. Three compartments on each side of trailer, each with a door, and all located forward of rear-wheel-housing.
b. One compartment on each side of trailer, each with door, and located to the rear of the rear-wheel-housing.
c. One transverse boxed-in compartment located directly under the aerial ladder bed. This compartment shall provide storage for long handle equipment, and shall open into the upper area in side compartments forward of rear wheels. The approximate size of this compartment shall be 100 inches wide, 54 inches long, and 7 inches high.

* 3.10.14.1 Compartment construction. The maximum cubic feet of compartment space shall be furnished on each side of the truck. Manufacturer's shall provide diagrams depicting compartment design and layout. All compartments shall be fabricated from aluminum and be at least .09 inches thick. Compartment floors will be integral with the compartment, be constructed of aluminum and be the sweep out type. Compartment door openings are to be framed by flanging edges to form a channel. Compartment doors will be flush mounted with double panel construction of aluminum. Outer panel door edges will be framed in a channel design to contain an inner pan. The interior of double panel doors will be coated to inhibit corrosion. A heavy duty automotive type extruded rubber moulding shall be installed on interior door pan and exterior edges to ensure a weatherproof compartment. All compartment doors will have polished stainless steel continuous piano type hinges. Hinge pin diameter will be a minimum of 3/16 inches and stainless steel material. Doors will be equipped with Eberhard M206, or equal, two-point latches that close on a single striker. Door handles will be the bent "D" ring, polished stainless steel type. The free door, of double equipped compartments will have a latch also. Cleveland type door holders are to be installed on each door. Switches for automatic compartment lights are to be installed at the hinge side of door. Drip moulding will be installed above all compartment door openings. The aluminum treadplate walkways over the fender side compartments shall have rolled over edges (FRONT-REAR-OUTSIDE EDGES) and formed out to provide drip protection, with inner body edge being rolled upward a minimum of 1.5 inches. A compartment shall be provided (MID-SHIP MOUNTED) and contain a cascade system for recharging self contained breathing apparatus. The cascade system shall meet NIOSH and ANSI standards with the capacity to recharge a minimum of 25 thirty minute air bottles. The unit shall have a fill station on the left side, consisting of the following: pressure gages for each cylinder, one master shut off gage for the fill line; and one six foot hose for filling the air bottles.

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3.10.15 Tractor-trailer coupling. The tractor-trailer coupling device connecting the two vehicles shall be designed, constructed, and installed in accordance with the fire truck manufacturer's requirements for safe and proper operation of the complete unit as intended by the fire service. The coupling device shall conform to applicable DoT Federal Motor Carrier Safety Regulations, section 393.70.

3.10.16 Aerial ladder two-way communication system. A two-way intercom system shall be provided at the aerial ladder platform operator's position and the outer fly section to allow continuous communication between ladder operator and firefighters. The intercom unit shall be similar or equal to model IC-9 manufactured by Ratelco, Inc. The system shall be powered by chassis battery system, shall have at least 18 watts output, and shall be weatherproof designed. The firefighter aloft at the fly section shall have no controls to operate ("Hands-off" communication).

3.11 Treatment and painting. Vehicle surfaces shall be cleaned, and treated in accordance with MIL-T-704, type A. The exterior surface to be painted shall be primed with a zinc chromate epoxy applied to a dry film thickness of 1.5 - 2.5 mils. The primer used shall be compatible with the final paint system. The finish paint shall be polyurethane enamel or equal, applied to a dry film thickness of 1.8 - 2.2 mils. The finish color shall be lime yellow gloss, number 13670 of FED-STD-595 and shall be applied to the exterior surface of the truck cab, chassis, and body. The roof and upper part of the cab, to a line at the bottom of the windshield, shall be painted white, color chip No. 17886 of FED-STD-595. A white reflective glow type, horizontal "accent" band, 10 inches wide, shall be installed along the sides, the full length of the truck. The top of the band shall align approximately with the top of the wheel well opening. The interior surface of the equipment compartments shall be painted with spatter type paint to reduce problems in repainting due to chipping from equipment storage. Running boards, compartment tops, steps and turntable work platform shall be of aluminum color. Cab interior shall be painted two coats semigloss enamel conforming to TT-E-529 and color chip No. 24525 of FED-STD-595.

3.12 Identification marking. Identification marking shall be in accordance with MIL-STD-1223 as required for the Navy.

* 3.13 Radio interference suppression. The vehicle shall be suppressed to limit electromagnetic radiation in accordance with SAE J551.

* 3.14 Identification plate. The contracting officer will furnish to the Government inspector the required identification plates. The contractor will be required to stamp the necessary data in the blank spaces thereon and securely affix said plates in a conspicuous place on each unit, assembly or

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subassembly, and parts as directed by the Government inspector. Nonferrous screws, rivets, or bolts of not less than 1/8-inch in diameter shall be used to affix the plates. Nomenclature shall be "TRUCK, FIREFIGHTING, AERIAL LADDER, 100 FOOT LENGTH."

* 3.15 Instruction plates. The truck shall be equipped with instruction plates suitably located, describing any special or important procedures to be followed in operating and servicing the equipment. Plates shall be of a material which will last and remain legible for the life of the equipment. Plates shall be securely affixed to the equipment with nonferrous screws or bolts of not less than 1/8-inch diameter.

3.16 Ease of maintenance. The truck shall be so designed and constructed that normal adjustments, repair, and overhaul can be readily accomplished by means of general purpose tools with a minimum removal or distrubance of other elements of the unit. Ease of maintenance provisions shall insure operating clearances for facilitating maintenance and servicing. Where possible, intricate locking devices, controls, and threaded fastenings that can be easily over-torqued by operators lacking feel through thick gloves or numbness, shall be avoided. Covers or plates that must be removed for component adjustment or for parts removal shall be equipped with substantial, quick-disconnect fastenings.

* 3.17 Lubrication. Means for lubrication shall be in accordance with the manufacturer's standard practice. The lubricating points shall be easily visible and accessible. Hydraulic lubrication fittings shall be in accordance with SAE J534. Where use of high-pressure lubricating equipment, 1,000 psi or higher, will damage grease seals or other parts, a suitable warning shall be affixed to the equipment in a conspicuous location. The unit shall be lubricated prior to delivery with type of lubricant specified in the operator's manual and grade of lubricant recommended for ambient temperature at the delivery point. The unit shall be conspicuously tagged to identify the lubricants and their temperature range.

* 3.18 Servicing and adjusting. Prior to acceptance of the aerial ladder truck by the Government, the contractor shall service and adjust the aerial ladder truck for immediate operational use as required in the operator's manual. The servicing and adjusting shall include at least the following:

a. Inflation of all tires.
b. Adjustment of brakes.
c. Proper functioning of all lighting and electrical systems.
d. Wheel alignment.
e. Adjustment of engine to include tune-up.
f. Complete lubrication with grades of lubricants recommended for ambient temperature at the delivery point.

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g. Cooling system filled to capacity with a clean solution of equal parts by volume of water and antifreeze (ethylene glycol).

The aerial ladder truck shall be conspicuously tagged to identify the lubricants and their temperature range.

3.19 Workmanship.

3.19.1 Metal fabrication. The metal used in fabrication shall be free from kinks, sharp bends, and other conditions which would be deleterious to the finished product. Manufacturing processes shall not reduce the strength of the metal to a value less than intended by the design. Shearing and chipping shall be done neatly and accurately. All bends shall be made by controlled means to insure uniformity of size and shape.

3.19.2 Bolted connections. Boltholes shall be accurately punched or drilled and shall have the burrs removed. Washers or lockwashers shall be provided in accordance with good commercial practice, and all bolts, nuts, and screws shall be tight.

3.19.3 Riveted connections. Rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall be driven with pressure tools and shall completely fill the holes. Rivet heads, when not countersunk or flattened, shall be of approved shape and of uniform size for the same diameter of rivet. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member.

3.19.4 Welding. Welding procedures shall be in accordance with a nationally recognized welding code. The surface of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the parts connected by the welds. Welds shall transmit stress without permanent deformation or failure when the parts connected by the weld are subjected to proof and service loadings.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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* 4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements specified herein and in applicable referenced documents.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1).
b. Quality conformance inspection (see 4.2.2).

4.2.1 First article inspection. The first article inspection shall be performed on one truck when a first article is required (see 3.2 and 6.1). This inspection shall include the examination of 4.3 and the tests of 4.4. The first article may be either a first production item or a standard production item from the supplier's current inventory provided the item meets the requirements of the specification and is representative of the design, construction, and manufacturing technique applicable to the remaining items to be furnished under the contract.

4.2.2 Quality conformance inspection. The quality conformance inspection shall include the examination of 4.3, the tests of 4.5, and the packaging inspection of 4.6.

4.3 Examination. Each truck shall be examined for compliance with the requirements specified in section 3 of this specification. Any redesign or modification of the contractor's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirements or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

4.4 First article tests. When first article is required, the first article shall be tested as specified in 4.4.1 through 4.4.5. The first article tests shall be performed by the contractor under the direction and in the presence of Government representative. Failure to pass any phase of the required test shall be cause for the Government to refuse acceptance of the truck until corrective action has been taken.

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* 4.4.1 Operational tests. The fully equipped truck, loaded as specified in 3.7.4, shall be tested as follows:

a. Ten miles over paved roads at maximum traffic speeds up to 55 mph to verify conformance to the requirements of 3.6d.
b. Gradeability test to verify conformance to the requirements of 3.6a.
c. Acceleration test to verify conformance to the requirements of 3.6b and 3.6c.
d. Service brake tests to verify conformance to the requirements of 3.6e.
e. Parking brake test to verify conformance to the requirements of 3.6f.

* 4.4.2 Aerial ladder tests. A structural strength test or the aerial ladder shall be conducted in accordance with NFPA 1904. In lieu of testing, an approved third party inspection certificate accompanied by the aerial ladder manufacturer's certification that the aerial ladder furnished has been previously tested to comply with NFPA 1904, is acceptable. In addition, an operational test of the aerial ladder shall be conducted. A complete cycle of aerial ladder operation shall be conducted. A complete cycle of aerial ladder operation shall be carried out after starting the engine, setting the outrigger(s), and transmitting power to the ladder. Elevating the ladder out of the bed, rotating ladder 90° and extending ladder to full specified height shall be completed smoothly and without undue vibration in 60 seconds or less. The ladder shall be retracted, the turntable rotation completed and the ladder lowered to its bed after which a thorough inspection shall be made of all moving parts with special attention being given to the security and adjustment of the ladder cables. The test shall include successful operation of all ladder controls, safety devices, auxiliary power system and ground outrigger devices.

* 4.4.2.1 Ladder pipe tests. The aerial ladder and water system shall be tested for compliance to requirements specified in paragraphs 3.10.10 and 3.10.10.1.

4.4.3 Ground ladders test. Each ground ladder specified in 3.10.13 shall be tested in accordance with NFPA 1931. In lieu of testing, inspection accompanied by ladder manufacturer's certification that the ladders have been factory tested and comply with NFPA 1931 is acceptable.

4.4.4 Hydraulic system tests. The hydraulic system shall be checked for conformance to 3.9 through 3.9.5. With pump engaged and operating at recommended speed, check hydraulic pressure to see that it is within limits recommended by the manufacturer.

* 4.4.5 Engine and cooling system tests. A high and low temperature operating test shall be performed to verify engine and cooling system requirements of paragraphs 3.5 and 3.8.3.1. Certification of these requirements can be furnished in lieu of hot and cold room tests.

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4.4.6 Production sample. Upon acceptance of the first article, the first article shall remain at the manufacturing facility as a production sample, and shall be reconditioned, including replacement of abnormally worn parts and paint touch-up or repainting, prior to delivery to enable it to be accepted as a contract item. The contractor shall maintain the first article in a serviceable condition for the duration of the contract.

4.5 Quality conformance tests. Each production truck shall be tested as specified in paragraphs 4.4.2 through 4.4.5. Failure to conform satisfactorily to any phase of the tests shall be cause for rejection.

4.6 Packaging inspection. The preservation, packing, and marking of the item shall be inspected to verify conformance to the requirements of section 5.

5. PACKAGING

5.1 Vehicle processing. The equipment shall be preserved and packaged in accordance with the contractor's standard practice. When specified (see 6.1), equipment shall be preserved and packaged in accordance with the mobile requirements of MIL-STD-281 with the level of preservation and packaging specified (see 6.1).

6. NOTES

6.1 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. When first article is required for inspection and approval (see 3.2, 4.2.1, and 6.3).
c. When vehicle shall be preserved and packaged in accordance with MIL-STD-281 with the level of preservation and packaging required (see 5.1).

* 6.2 Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL), incorporated into the contract. When the provisions of DOD FAR Supplement, Part 27, Sub-Part 27.475-1 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data shall be delivered by the contractor in accordance with the contract or purchase order requirements.

* 6.3 First article. When a first article inspection is required, the item will be tested and should be a first production item consisting of one complete truck or it may be a standard production item from the contractor's current inventory as specified in 4.2.1. The first article should consist of one complete unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

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6.4 Performance prediction. Completed copies of SAE J688, Truck Ability Prediction Procedure Form and computation for low speed requirement for each vehicle model furnished under contract should be submitted as specified in the contract. The SAE Work Sheet Item 1 should include vehicle model number, wheel base, and engine model number. Unless other conditions are cited in the contract, computations should be made for normal atmospheric pressure, normal ambient air temperature, and still dry air. The factors to be used in predicting truck ability are established as follows for the corresponding SAE Truck Ability Prediction Procedure Tables:

Table 1	-	Tire Factor. This factor must relate to the size of tires furnished by the contractor in accordance with this specification.
Table 2	-	Altitude Factor. 1.00
Table 3	-	Rolling Factor. 1.613
Table 4	-	Area Factor. Vehicle frontal dimensions.
Table 5	-	Velocity Factor. 250.00
Table 6	-	Altitude Factor. 1.00
Table 7	-	Chassis Friction Horsepower. Use applicable power unit gvw (to nearest, higher, 1,000 pounds) and the engine rpm (to nearest 100 revolutions) which is required for 55 mph geared speed. For gvw and engine speed beyond this table, factors shall be extrapolated.
Table 8	-	Grade Factor. 0.75
Table 8A	-	Correction Factor. Not required.
Table 9	-	Road Factor. 0.0

* 6.5 Subject term (key word) listing.

Diesel engine
Fire fighting vehicle
Fire protection
Fire pump

6.6 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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